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Display device

Abstract

A display device is disclosed. The display device of the present disclosure comprises: a flexible display panel; a roller that is elongated and rolls up or unrolls the display panel; a housing in which the roller is located and having an opening through which the display panel passes; and a door assembly which is coupled to the housing adjacent to the opening, and opens and closes the opening, wherein the door assembly comprises: a door for opening or closing the opening; and a suspender positioned between the housing and the door and having one end fixed to one of the housing and the door, and the other end movably coupled to the other one.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application

No. PCT/KR2020/003014, filed on Mar. 3, 2020, the contents of which are hereby incorporated by reference herein its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a display device.

BACKGROUND ART

(3) As the information society develops, the demand for display devices is also increasing in various forms. In response to this, various display devices such as Liquid Crystal Display Device (LCD), Plasma Display Panel (PDP), Electroluminescent Display (ELD), and Vacuum Fluorescent Display (VFD) have been researched and used in recent years.

(4) Among them, a display device using an organic light emitting diode (OLED) has superior luminance characteristics and viewing angle characteristics compared to a liquid crystal display device, and can be implemented in an ultra-thin shape as it does not require a backlight unit.

(5) In addition, a flexible display panel can be bent or wound on a roller. By using the flexible display panel, it is possible to implement a display device that is roll out from a roller or wound on a roller. A lot of research has been done on a structure for winding or unwinding a flexible display panel on/from a roller.

DISCLOSURE

Technical Problem

(6) An object of the present disclosure is to solve the above and other problems.

(7) Another object of the present disclosure may be to provide a display device capable of preventing sagging of a door that opens and closes an opening through which a display panel passes.

(8) Another object of the present disclosure may be to provide a display device capable of preventing sagging of a door continuously, in addition to duration when the door moves.

(9) Another object of the present disclosure may be to provide a display device capable of preventing a step difference, between a door and a housing adjacent thereto, that is generated according to a sagging of the door.

Technical Solution

(10) According to an aspect of the present disclosure, there is provided a display device including: a flexible display panel; a roller which extends long, and around which the display panel is wound, or from which the display panel is unwound; a housing which has the roller located therein, and has an opening through which the display panel passes; and a door assembly which is coupled to the housing while being adjacent to the opening, and opens and closes the opening; wherein the door assembly includes: a door which opens and closes the opening; and a suspender located between the housing and the door, the suspender that has one end fixed to any one of the housing and the door, and has the other end movably coupled to the other of the housing and the door.

Advantageous Effects

(11) The effect of the display device according to the present disclosure will be described as follows.

(12) According to at least one of embodiments of the present disclosure, it is possible to provide a display device capable of preventing sagging of a door that opens and closes an opening through which a display panel passes.

(13) According to at least one of embodiments of the present disclosure, it is possible to a display device capable of preventing sagging of a door continuously, in addition to duration when the door moves.

(14) According to at least one of embodiments of the present disclosure, it is possible to provide a display device capable of preventing a step difference, between a door and a housing adjacent thereto, that is generated according to a sagging of the door.

(15) Further scope of applicability of the present disclosure will become apparent from the following detailed description. However, it should be understood that the detailed description and

specific embodiments such as preferred embodiments of the present disclosure are given by way of example only, since various changes and modifications within the spirit and scope of the present disclosure may be clearly understood by those skilled in the art.

Description

DESCRIPTION OF DRAWINGS

(1) FIGS. **1** to **33** are diagrams illustrating examples of a display device according to embodiments of the present disclosure.

MODE FOR DISCLOSURE

(2) Description will now be given in detail according to exemplary embodiments disclosed herein, with reference to the accompanying drawings. For the sake of brief description with reference to the drawings, the same or equivalent components may be denoted by the same reference numbers, and description thereof will not be repeated.

(3) In general, suffixes such as “module” and “unit” may be used to refer to elements or components. Use of such suffixes herein is merely intended to facilitate description of the specification, and the suffixes do not have any special meaning or function.

(4) In the present disclosure, that which is well known to one of ordinary skill in the relevant art has generally been omitted for the sake of brevity. The accompanying drawings are used to assist in easy understanding of various technical features and it should be understood that the embodiments presented herein are not limited by the accompanying drawings. As such, the present disclosure should be construed to extend to any alterations, equivalents and substitutes in addition to those which are particularly set out in the accompanying drawings.

(5) It will be understood that although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

(6) It will be understood that when an element is referred to as being “connected with” another element, there may be intervening elements present. In contrast, it will be understood that when an element is referred to as being “directly connected with” another element, there are no intervening elements present.

(7) A singular representation may include a plural representation unless context clearly indicates otherwise.

(8) In the following description, even if the embodiment is described with reference to specific drawings, if necessary, reference numerals not appearing in the specific drawings may be referred to, and reference numerals not appearing in the specific drawings are used in a case where the above reference numerals appear in the other figures.

(9) Referring to FIG. **1**, a display device **100** may include a display unit **20** and a housing **30**. The housing **30** may have an internal space. At least a portion of the display unit **20** may be located inside the housing **30**. At least a portion of the display unit **20** may be located outside the housing **30**. The display unit **20** may display a screen.

(10) A direction parallel to the length direction of the housing **30** may be referred to as a first direction DR**1** or a left-right direction LR. A direction parallel to the height direction of the housing **30** may be referred to as a second direction DR**2** or an up-down direction UD. A direction perpendicular to the left-right direction LR and the up-down direction UD may be referred to as a third direction DR**3** or a front-rear direction FR. A direction in which the display unit **20** displays a screen may be referred to as a forward direction, and a direction opposite to this may be referred to as a rearward direction.

(11) Hereinafter, a display deployment type in which the display unit **20** is deployed while descending from a lower plate of the housing **30** with respect to the display device **100** and is rolled

while ascending to the lower plate of the housing **30** will be described as an example, but the display deployment type of the display device **100** applicable to the present disclosure is not limited thereto. That is, a display deployment type in which the display unit **20** is deployed while ascending from an upper plate of the housing **30** with respect to the display device **100** and is rolled while descending to the upper plate of the housing **30** may also be applied to the present disclosure.

(12) Referring to FIG. 2, the entire display unit **20** may be located inside the housing **30**. At least a portion of the display unit **20** may be located outside the housing **30**. The extent to which the display unit **20** is exposed to the outside of the housing **30** may be adjusted as necessary.

(13) Referring to FIG. 3, the display unit **20** may include a display panel **21** and a plate **23**. The display panel **21** may be flexible. For example, the display panel **21** may be an organic light emitting display (OLED).

(14) The display panel **21** may have a front surface for displaying an image. The display panel **21** may have a rear surface opposite to the front surface. The front surface of the display panel **21** may be covered with a light-transmitting material. For example, the light-transmitting material may be a synthetic resin or a film.

(15) The plate **23** may be coupled, fastened, or attached to the rear surface of the display panel **21**. The plate **23** may include a metal material. The plate **23** may be referred to as a module cover **23**, a cover **23**, a display panel cover **23**, a panel cover **23**, or an apron **23**.

(16) Referring to FIG. 4, the plate **23** may include a plurality of segments **23c**. A magnet **24** may be located inside a recess **26** of the segment **23c**. The recess **26** may be located on a surface of the segment **23c** facing the display panel **21**. The recess **26** may be located in the front surface of each segment **23c**. Since the magnet **24** is received inside the recess **26**, the magnet **24** may not protrude out of the segment **23c**. The display panel **21** may be flat without being crumpled even when it comes into contact with the segment **23c**.

(17) Referring to FIG. 5, a bead **27** may be formed on the upper surface of the segment **23b**. The bead **27** may have a shape recessed to the inside of the segment **23b**. For example, the bead **27** may be formed by pressing the segment **23b**. A plurality of beads **27** may be formed in the segment **23b**. The plurality of beads **27** may be spaced apart from each other. The bead **27** may improve the rigidity of the segment **23b**. The bead **27** may prevent the shape of the segment **23b** from being deformed from an external impact. The segment **23b** may be fixed to the rearward direction of the display panel **21** by an adhesive member **22**. For example, the adhesive member **22** may be a double-sided tape.

(18) Referring to FIG. 6, a source PCB **21b** may be located in the lower side of the module cover **23**. When the source PCB **21b** is rolled down or rolled up, its position may change with the movement of the module cover **23**. A Flexible Flat Cable (FFC) cable **21a** may be located in the center of the module cover **23** based on a first direction. The FFC cable **21a** may be located in opposite ends of the module cover **23** based on the first direction.

(19) Referring to FIG. 7, the segment **23d** may include a recess portion **28** recessed to the inside of the segment **23d**. The recess portion **28** may form a space between the display panel **21** and the module cover **23**. The FFC cable **21a** may be accommodated in the space formed by the recess portion **28**. In addition, the recess portion **28** may improve the rigidity of the segment **23d**. The bead **27** may be located on the segment **23d** excluding a portion where the recess portion **28** is located. The position of the recess portion **28** may not overlap with the position of the bead **27** to prevent the segment **23d** from weakening.

(20) Referring to FIG. 8, in the segment **23e**, a penetrating portion **29** may be located in a central portion of the segment **23e** based on the first direction. The penetrating portion **29** may penetrate the central portion of the segment **23e** in the second direction. That is, the penetrating portion **29** may be a hole located in the segment **23e**. The penetrating portion **29** may be a portion in which the FFC cable **21a** is located. Since the penetrating portion **29** is formed in the segment **23e**, the thickness of the segment **23e** can be reduced in comparison with a case where the FFC cable **21a** is

located in the recess portion **28** (refer to FIG. 7). The bead **27** may be located in the segment **23e** excluding a portion where the penetrating portion **29** is located. The position of the penetrating portion **29** may not overlap the position of the bead **27** in order to prevent the segment **23e** from weakening.

(21) Referring to FIG. 9, a top case **60** may cover the display panel **21** and the module cover **23** as well as the source PCB **21b** and an upper bar **201**. One surface of the upper bar **201** may be coupled to the rear surface of the module cover **23**, and the other surface may be coupled to the source PCB **21b**. The upper bar **201** may be fixed to the module cover **23** to support the source PCB **21b**.

(22) The lower end of the FFC cable **21a** may be connected to a timing controller board **11** (refer to FIG. 10) inside a panel roller **10** (refer to FIG. 10). The FFC cable **21a** may be wound around or unwound from the panel roller **10** together with the display unit **20**.

(23) A portion of the FFC cable **21a** may be located between the display panel **21** and the module cover **23**. A portion of the FFC cable **21a** located between the display panel **21** and the module cover **23** may be referred to as a first portion **21a1**. The first portion **21a1** may be located in the recess portion **28** formed by the plurality of segments **23d**. Alternatively, the first portion **21a1** may be received in the recess portion **28** formed by the plurality of segments **23d**.

(24) A portion of the FFC cable **21a** may penetrate the segment **23f**. A portion of the FFC cable **21a** that penetrates the segment **23f** may be referred to as a second portion **21a2**. The segment **23f** may include a first hole **23fh1** formed on the front surface and a second hole **23fh2** formed on the rear surface. The first hole **23fh1** and the second hole **23fh2** may be interconnected to form one hole **23fh**. The hole **23fh** may penetrate the segment **23f** in a third direction. The second portion **21a2** may pass through the hole **23fh**. The hole **23fh** may be referred to as a connection hole **23fh**.

(25) An upper end of the FFC cable **21a** may be electrically connected to the source PCB **21b**. A portion of the FFC cable **21a** may be located on the rear surface of the module cover **23**. A portion of the FFC cable **21a** located on the rear surface of the module cover **23** may be referred to as a third portion **21a3**. The third portion **21a3** may be electrically connected to the source PCB **21b**.

(26) The third portion **21a3** may be covered by the top case **60**. Accordingly, the third portion **21a3** may not be exposed to the outside.

(27) Referring to FIG. 10, the FFC cable **21a** may be connected to the timing controller board **11** mounted on the panel roller **10**. A through hole **12** may be formed in the panel roller **10**, and the FFC cable **21a** may be connected to the timing controller board **11** through the through hole **12**.

(28) The through hole **12** may be located in one side of the panel roller **10** and may penetrate the outer circumference portion of the panel roller **10**. The FFC cable **21a** may be connected to one side of the timing controller board **11** through the through hole **12**.

(29) Even if the FFC cable **21a** is located on the outer circumference of the panel roller **10**, the connection to the timing controller board **11** may be maintained due to the through hole **12**.

Accordingly, the FFC cable **21a** may not be twisted by rotating together with the panel roller **10**.

(30) A portion of the FFC cable **21a** may be wound around the panel roller **10**. A portion of the FFC cable **21a** wound around the panel roller **10** may be referred to as a fourth portion **23a4**. The fourth portion **23a4** may be in contact with the circumferential surface of the panel roller **10**.

(31) A portion of the FFC cable **21a** may pass through the through hole **12**. A portion of the FFC cable **21a** passing through the through hole **12** may be referred to as a fifth portion **23a5**.

(32) A lower end of the FFC cable **21a** may be electrically connected to the timing controller board **11**. A portion of the FFC cable **21a** may be located inside the panel roller **10**. A portion of the FFC cable **21a** located inside the panel roller **10** may be referred to as a sixth portion **23a6**. The sixth portion **23a6** may be electrically connected to the timing controller board **11**.

(33) Referring to FIG. 11, the display panel **21** may be connected to the roller **10**. The display panel **21** may be wound around or unwound from the roller **10**. The display panel **21** may be electrically connected to the plurality of source PCBs **21b**. The plurality of source PCBs **21b** may be spaced

apart from each other.

(34) A source chip on film (COF) **21m** may connect the display panel **21** and the source PCB **21b**. The source COF **21m** may be located in the lower side of the display panel **21**. The roller **10** may include a first part **13** and a second part **14**. The first part **13** and the second part **14** may be fastened by a screw. The timing controller board **11** may be mounted inside the roller **10**.

(35) The source PCB **21b** may be electrically connected to the timing controller board **11**. The timing controller board **11** may transmit digital video data and a timing control signal to the source PCB **21b**.

(36) A cable **21p** may electrically connect the source PCB **21b** and the timing controller board **11**. For example, the cable **21p** may be a flexible flat cable (FFC). The cable **21p** may pass through the hole **21n**. The hole **21n** may be formed in a seating portion **15** or in the first part **13**. The cable **21p** may be located between the display panel **21** and the second part **14**.

(37) The seating portion **15** may be formed in the outer circumference of the first part **13**. The seating portion **15** may be formed by stepping a portion of the outer circumference of the first part **13**. The seating portion **15** may form a space B. When the display unit **20** is wound around the panel roller **10**, the source PCB **21b** may be received in the seating portion **15**. Since the source PCB **21b** is received in the seating portion **15**, it may not be bent or crooked, and durability may be improved.

(38) Referring to FIGS. **12** and **13**, a door assembly **40** may be installed inside the housing **30** while being adjacent to the lower plate of the housing **30**. The door assembly **40** may be coupled to the housing **30** while being adjacent to an opening **30P** of the lower plate of the housing **30**. Here, the opening **30P** of the lower plate of the housing **30** may be an opening through which the display panel **21** passes.

(39) A door **41** may cover the opening **30P** of the lower plate of the housing **30**. The opening **30P** of the lower plate of the housing **30** may extend from the lower plate of the housing **30** in a length direction of the housing **30**. The door **41** may be a plate that extends long. The door **41** may move while sliding in the front-rear direction of the housing **30**. A holder **42** may be coupled to the door **41**. One side of the holder **42** may be fixed to the upper surface of the door **41**. In this case, the holder **42** may be located adjacent to the right end in the length direction of the door **41**. The holder **42** may be referred to as a door holder **42**.

(40) The holder **42** may be coupled to a slider **43**. The other side of the holder **42** may be fixed to the slider **43**. The slider **43** may be movable on a rail **44**. The rail **44** may be fixed to the upper side of the lower plate of the housing **30**. The rail **44** may include slots **44S** extending in the front-rear direction of the housing **30**. The slider **43** may be inserted into the slot **44S** and may move in the extending direction of the slot **44S**. A rack gear **43R** may be formed in the upper surface of the slider **43**.

(41) A shaft gear **45** may include a first gear **45a** and a second gear **45b**. The first gear **45b** may be engaged with the rack gear **43R**, and the second gear **45a** may be fixed to the first gear **45b**. For example, the first gear **45b** and the second gear **45a** may be formed as one body. A shaft **50** may be inserted into the first gear **45a** and the second gear **45b**. The shaft **50**, the first gear **45a**, and the second gear **45b** may rotate simultaneously.

(42) A motor **49** may provide rotational force. For example, the motor **49** may provide a torque of 4 kg.Math.m. The motor **49** may be installed inside the housing **30**. The motor **49** may be connected to a worm **48**. When the motor **49** rotates, the worm **48** may rotate. A worm gear **47** may be engaged with the worm **48** connected to the motor **49**. The worm gear **47** may be connected to a torque limiter **46**. For example, the torque limiter **46** may limit the torque exceeding 2 kg.Math.m. The torque limiter **46** may be connected to the shaft gear **45**. For example, the torque limiter **46** may be engaged with the second gear **45a** of the shaft gear **45**.

(43) When the motor **49** rotates, the worm **48** may provide power to the worm gear **47**. The worm gear **47** may provide power to the shaft gear **45** through the torque limiter **46**. Accordingly, in a

situation in which the driving of the door **41** is prevented, a damage to mechanism and an injury due to user's hand being caught can be prevented.

(44) Referring to FIGS. **14** and **15**, a first shaft gear **45** may be installed adjacent to the motor **49**, and may receive driving force from the motor **49**. The first shaft gear **45** may be fixed to a first shaft **50a** to provide rotational force to the first shaft **50a**. The first shaft **50a** may be extend long in the left-right direction of the housing **30**, and may be hold by a holder **60H**.

(45) A second shaft gear **45'** may be installed in the other side opposite to one side of the housing **30** where the motor **49** is located, and be fixed to a second shaft **50b**. As the second shaft **50b** rotates, the second shaft gear **45'** may be rotated. The first shaft **50a** and the second shaft **50b** may be connected or coupled by a joint (no reference numeral) located inside the holder **60H**. Meanwhile, the number of shafts **50** may be two or more. When the number of shafts **50** is increased to n , the number of joints may be increased to $n-1$ correspondingly.

(46) The first shaft gear **45** may move the first slider **43** in the front-rear direction of the housing **30**, and the second shaft gear **45'** may move a second slider **43'** in the front-rear direction of the housing **30** while opening and closing the door **41**.

(47) Accordingly, the door **41** may be opened and closed while the shafts **50** rotate without being caught in the holder **60H** or causing noise and/or vibration.

(48) Referring to FIG. **16**, an example in which the door **41** installed in the lower plate of the housing **30** is opened by rotation of the shaft **50** can be seen. Referring to FIG. **17**, an example in which the display unit **20** descends between the opening **30P** of the lower plate opened by the movement of the door **41** can be seen.

(49) Referring to FIG. **18**, the door **41** may extend in the length direction of the housing **30**, that is, in the left-right direction. In addition, one side of the door **41** may be coupled to one side of the housing **30** through the slider **43** and the rail **44** based on the length direction of the door **41**, and the other side of the door **41** may be coupled to the other side of the housing **30** through a slider **43'** and a rail **44'**.

(50) Here, when only both sides of the elongated door **41** are coupled to the housing **30**, a portion adjacent to the center of the door **41** may sag downward due to its own weight. In addition, when the door **41** is sagged, a step difference may occur with respect to the housing **30**, thereby reducing the sense of unity of a device.

(51) Accordingly, it is necessary to prevent the door **41** from sagging by holding the door **41** at between both sides of the door **41** based on the length direction of the door **41**.

(52) Referring to FIGS. **19** and **20**, a suspender SU may be located between the housing **30** and the door **41**. For example, the suspender SU may be located between the housing **30** and the door **41** based on the up-down direction of the housing **30**. The suspender SU may include a plurality of suspenders spaced apart from each other in the length direction of the housing **30**. One end of the suspender SU may be fixed to any one of the housing **30** and the door **41**. The other end of the suspender SU may be movably coupled to the other one of the housing **30** and the door **41**.

(53) The suspender SU may include a pin **90** that extends long. The length direction of the pin **90** may be parallel to the up-down direction of the housing **30**. In this case, the door assembly **40** may include a fastening part **71** and a sliding part **80**. The fastening part **71** may be formed in any one of the housing **30** and the door **41**. The sliding part **80** may be formed in the other one of the housing **30** and the door **41**.

(54) One end of the fastening part **71** may be fixed to the pin **90**. For example, the pin **90** may be formed in the shape of a long bolt, and the pin **90** may be screwed to the fastening part **71**. For example, the fastening part **71** may be formed in a portion of the housing **30** facing the door **41** in the up-down direction.

(55) The sliding part **80** may opposite the fastening part **71** with respect to the pin **90**. That is, the pin **90** may be located between the fastening part **71** and the sliding part **80** in the up-down direction. The sliding part **80** may be movably coupled to the other end of the pin **90**. For example,

the sliding part **80** may be formed in a portion of the door **41** facing the housing **30** in the up-down direction. Thus, the pin **90** that has one end fixed to the fastening part **71** and has a fixed position may slide with respect to the moving door **41**.

(56) The sliding part **80** may include a slot SL through which the pin **90** passes, and which extends long in a direction intersecting with the length direction of the pin **90**. For example, the length direction of the slot SL may be parallel to a direction in which the door **41** moves, i.e., the front-rear direction of the housing **30**.

(57) The sliding part **80** may include a first part **81** and a second part **82**. The first part **81** may be formed by penetrating the sliding part **80** in the up-down direction. The second part **82** may protrude from the inner surface of the first part **81**, and form a slot SL therein. The first part **81** and the second part **82** may be formed to have a step difference from each other, and may extend in the front-rear direction. The first part **81** and the second part **82** may be formed in a closed-loop shape, thereby preventing the pin **90** inserted into the slot SL based on the front-rear direction from escaping to the outside the slot SL.

(58) The pin **90** may include a body **91**, a neck **92**, and a head **93**. One end of the body **91** may be fixed to the fastening part **71**. For example, a thread is formed in one end of the body **91**, so that the body **91** may be screwed to the fastening part **71**. In this case, the fastening part **71** may be referred to as a screw hole. The neck **92** may be coupled to the other end of the body **91** and inserted into the slot SL. The body **91** and the neck **92** may be formed in a cylindrical shape. For example, the neck **92** may have a larger diameter than the diameter of the body **91**. The head **93** may oppose the body **91** with respect to the neck **92**. The head **93** may be formed in a shape of screw head. For example, the head **93** may have a larger diameter than a gap of the slot SL.

(59) In this case, the outer surface of the neck **92** may be in contact with the inner surface of the second part **82**. That is, when the door **41** moves, the second part **82** may move in the front-rear direction with respect to the neck **92**. In other words, when the door **41** moves, the neck **92** may slide along the inner surface of the second part **82**.

(60) In addition, the head **93** may overlap with the second part **82** in the length direction of the pin **90**. That is, the upper surface of the head **93** may be in contact with the lower surface of the second part **82** to support the second part **82**. For example, the second part **82** may be located between the upper end and the lower end of the sliding part **80**, and the lower surface of the head **93** may be located in parallel with or in the upper side of the lower surface of the sliding part **80**.

(61) In addition, when the door **41** moves, the second part **82** may move in the front-rear direction with respect to the head **93**. In other words, when the door **41** moves, the head **93** may slide along the lower surface of the second part **82**.

(62) A sheet **82a** may be located between the second part **82** and the neck **92**. The sheet **82a** may be coupled, fastened, or attached to the inner surface of the second part **82**. The sheet **82a** may include a plastic material. For example, the sheet **82a** may include a material such as polycarbonate PC or polyoxymethylen POM.

(63) Accordingly, the sheet **82a** may prevent noise from being generated between the sliding operations of the neck **92** with respect to the second part **82**.

(64) Referring to FIG. 21, when the door **41** covers all of the opening **30P**, the pin **90** may be located adjacent to the inner rear surface of the sliding part **80** based on the front-rear direction of the housing **30**. As the door **41** moves in the direction in which the opening **30P** is opened, the pin **90** may be closer to the inner front surface of the sliding part **80** based on the front-rear direction of the housing **30**. That is, the sliding part **80** is moved in the front-rear direction with respect to the pin **90** inserted into the slot SL, and in response, the opening **30P** may be opened or closed or the degree of opening may be adjusted.

(65) In this case, the head **93** may continuously support the second part **82** of the sliding part **80**. That is, the door **41** may be movably coupled to the housing **30** by the pin **90**, thereby continuously preventing the door **41** from sagging due to its own weight. In addition, the housing **30** and the

door **41** share the same amount of sagging through the pin **90**, thereby preventing a step difference occurred between the two.

(66) A plurality of pins **90** may be provided, and a plurality of fastening parts **71** and sliding parts **80** may be provided corresponding to the number of pins **90**. In this case, the plurality of pins **90** may be located between both sides in the length direction of the door **41**, and may be spaced apart from each other. Accordingly, the door **41** may be prevented from sagging at a plurality of points by the plurality of pins **90**.

(67) Meanwhile, a brush **39** may be coupled to the housing **30** at a position adjacent to the opening **30P**. The brush **39** may include a plurality of comb teeth made of fiber or plastic. The brush **39** may prevent foreign substances such as dust from being flowed into the housing **30** through the opening **30P**.

(68) Referring to FIGS. **22** and **23**, a sliding part **80'** may be formed while being recessed outwardly from the inner surface of the door **41**. The sliding part **80'** may be spaced apart from the outer surface of the door **41**. Here, the inner surface of the door **41** may mean the upper surface of the door **41**, the outer surface of the door **41** may mean the lower surface of the door **41**, and the outer side of the door **41** may mean the lower side of the door **41**.

(69) The sliding part **80'** may include a first part **81'** and a second part **82'**. The first part **81'** may be formed while being recessed downward from the door **41**. The second part **82'** may protrude from the inner surface of the first part **81'** and form a slot SL therein. The first part **81'** and the second part **82'** may be formed to have a step difference from each other.

(70) One end of the body **91** may be fixed to the fastening part **71**, and the outer surface of the neck **92** inserted into the slot SL may be in contact with the inner surface of the second part **82'**. In this case, a sheet **82a'** may be located between the second part **82'** and the neck **92**. The head **93** may be inserted into a space located between the lower surface of the second part **82'** and the lower surface of the door **41**. The upper surface of the head **93** may support the lower surface of the second part **82'**.

(71) Accordingly, the door **41** may be movably coupled to the housing **30** by the pin **90**, thereby continuously preventing the door **41** from sagging due to its own weight. In addition, the head **93** of the pin **90** may not be exposed to the outside through the lower surface of the door **41**.

(72) Referring to FIGS. **24** and **25**, a fastening part **71''** may be formed in a portion of the door **41** facing the housing **30** in the up-down direction. In this case, one end of the pin **90** may be fixed to the fastening part **71''**. Here, one end of the pin **90** may mean a lower end of the pin **90**.

(73) A sliding part **80''** may opposite the fastening part **71''** with respect to the pin **90**. That is, the pin **90** may be located between the fastening part **71''** and the sliding part **80''** in the up-down direction. The sliding part **80''** may be coupled to the other end of the pin **90** to be movable. Here, the other end of the pin **90** may mean an upper end of the pin **90**. For example, the sliding part **80''** may be formed in a portion of the housing **30** facing the door **41** in the up-down direction.

(74) The sliding part **80''** may include a first part **81''** and a second part **82''**. The first part **81''** may be formed by penetrating the sliding part **80''** in the up-down direction. The second part **82''** may protrude from the inner surface of the first part **81''** and form a slot SL therein. The first part **81''** and the second part **82''** are formed to have a step difference from each other, and may extend in the front-rear direction. The first part **81''** and the second part **82''** may be formed in a closed-loop shape, thereby preventing the pin **90** inserted into the slot SL based on the front-rear direction from escaping to the outside of the slot SL.

(75) One end of the body **91** may be fixed to the fastening part **71**, and the outer surface of the neck **92** inserted into the slot SL may be in contact with the inner surface of the second part **82''**. In this case, a sheet **82a''** may be located between the second part **82''** and the neck **92**. The head **93** may overlap with the second part **82''** in the length direction of the pin **90**. That is, the upper surface of the second part **82''** may be in contact with the lower surface of the head **93** to support the lower surface of the head **93**.

(76) Referring to FIGS. 25 and 26, the pin 90 having one end fixed to the fastening part 71" may move together with the door 41 while sliding with respect to the sliding part 80". At this time, the lower surface of the head 93 may be continuously supported by the second part 82", thereby continuously preventing the door 41 to which the pin 90 is fixed from sagging due to its own weight. In addition, the housing 30 and the door 41 share the same amount of sagging through the pin 90, thereby preventing a step difference generated between the two.

(77) In addition, the fastening part 71" may be formed by penetrating the door 41 in the up-down direction. In this case, one end of the body 91 may be exposed to the outside through the lower surface of the door 41. Meanwhile, the fastening part 71" may be formed while being recessed outwardly or downwardly from the inner surface or upper surface of the door 41. In this case, one end of the body 91 may not be exposed to the outside through the lower surface of the door 41.

(78) Referring to FIGS. 27 and 28, the suspender SU may include a pin 94 that has one end fixed to the fastening part 71 and the other end having a magnet 95. The pin 94 may extend long in the up-down direction of the housing 30, and may be formed in a cylindrical shape as a whole. Here, the fastening part 71 may be formed in the housing 30, and the sliding part 80 may be formed in the door 41.

(79) The sliding part 80 may include a magnetic substance 85 facing the magnet 95 in the length direction of the pin 94. Accordingly, a magnetic attraction may act between the magnet 95 and the magnetic substance 85.

(80) Alternatively, a magnet may be provided in the sliding part 80, and a magnetic substance may be provided in the pin 94. In addition, a magnet may be provided in each of the pin 94 and the sliding part 80, and different types of poles (N pole-S pole) may be disposed in a surface on which the magnet provided in the pin 94 and the magnet provided in the sliding part 80 face each other. Meanwhile, the magnet may be provided as an electromagnet. Thus, magnetic attraction may act between the pin 94 and the sliding part 80 through various combinations of magnets or magnetic substances, but for a brief description, a case in which the magnet 95 is provided in the other end of the pin 94 and the magnetic substance 85 is provided in the sliding part 80 will be described.

(81) The magnet 95 may be formed in a ball shape. A shaft 941 may protrude from the inner surface of the pin 94 and be coupled to the magnet 95. In this case, while the magnet 95 is moved on the magnetic substance 85, the magnet 95 may be rotated about the shaft 941.

(82) The magnetic substance 85 may be provided in the inner surface or upper surface of the door 41. The magnetic substance 85 may be coupled, fastened, or attached to the door 41, or may be provided to the door 41 as one body. The magnetic substance 85 may include iron Fe.

(83) The magnetic substance 85 may extend as much as the door 41 moves while opening and closing the opening 30P. The magnetic substance 85 may be formed in an elongated plate shape. The magnetic substance 85 may be recessed from the upper surface to the lower side to form a groove 86 for guiding the movement of the magnet 95. That is, the magnet 95 may slide while being in contact with the groove 86.

(84) A sheet 86a may be located between the magnet 95 and the groove 86. The sheet 86a may be coupled, fastened, or attached to the groove 86. The sheet 86a may include a plastic material. For example, the sheet 86a may include a material such as polycarbonate (PC) or polyoxymethylen (POM).

(85) Accordingly, the sheet 86a may prevent noise from being generated between the sliding operations of the magnet 95 with respect to the groove 86.

(86) Referring to FIGS. 28 and 29, when the door 41 moves, the groove 86 may move in the front-rear direction with respect to the magnet 95. In other words, when the door 41 moves, the magnet 95 may slide along the groove 86 while maintaining magnetic coupling with the groove 86.

(87) That is, the door 41 is movably coupled to the housing 30 by the pin 94, thereby continuously preventing the door 41 from sagging due to its own weight. In addition, the housing 30 and the door 41 share the same amount of sagging through the pin 90, thereby preventing a step difference

generated between the two.

(88) Referring to FIG. 30, a magnet **95'** may be formed in the shape of a roller, i.e., an elongated cylinder. In this case, in comparison with a case of being formed in a ball shape, a contact area between the magnet **95'** and the magnetic substance **85** is increased, so that the magnetic coupling between the magnet **95'** and the groove **86** can be more stably implemented.

(89) Referring to FIG. 31, a magnetic substance **85'** may be located to correspond to a portion of the movement trajectory of the door **41** facing the magnet **95**. For example, in related to opening and closing of the opening **30P**, the door **41** may be located in a first position for completely closing the opening **30P**, a second position for completely opening the opening **30P**, and a third position for closing a part of the opening **30P**.

(90) In this case, the magnetic substance **85'** may be located in a first point, a second point, and a third point on the door **41**. Here, the first point may be located in a rearward direction than the second point and the third point, the second point may be located in a forward direction than the first point and the third point, and the third point may be located between the first point and the second point. When the door **41** is located in the first position, a magnetic substance **851'** located in the first position may face the magnet **95**. When the door **41** is located in the second position, a magnetic substance **852'** located in the second position may face the magnet **95**. When the door **41** is located in the third position, a magnetic substance **853'** located in the third position may face the magnet **95**.

(91) Accordingly, magnetic coupling between the magnet **95** and the magnetic substance **85'** is formed at a certain position such as the first to third positions to prevent the door **41** from sagging due to its own weight. In addition, while the door **41** is moving, a contact due to the magnetic coupling between the magnet **95** and the magnetic substance **85'** is not accomplished, thereby preventing noise caused by the contact between the two.

(92) Meanwhile, in order to prevent noise due to contact due to magnetic coupling between the magnet **95** and the magnetic substance **85'** at a certain position such as the first to third positions, a pad (not shown) formed of a material such as rubber, and silicone may be located on the magnetic substance **85'**.

(93) Referring to FIGS. 32 and 33, the suspender **SU** may include a rod **96** that has one end fixed to the housing **30** and extends long in the length direction of the door **41**. For example, one end of the rod **96** may be fixed to a left plate or right plate of the housing **30** through a fastening member **97** such as a screw.

(94) In this case, the door assembly **40** may include a holder **75** that is coupled to the door **41** and has a hole **75a** into which the rod **96** is inserted. A portion of the outer surface of the rod **96** may be in contact with the inner surface of the holder **75** defining the hole **75a**. The hole **75a** may extend long in the moving direction of the door **41**, i.e., in the front-rear direction. Thus, when the door **41** moves, the holder **75** may move in the front-rear direction with respect to the rod **96** having one end fixed to the housing **30**. In other words, when the door **41** moves, the rod **96** may slide along the inner surface of the holder **75** defining the hole **75a**.

(95) That is, the door **41** is movably coupled to the housing **30** by the rod **96**, thereby continuously preventing the door **41** from sagging due to its own weight. In addition, the housing **30** and the door **41** share the same amount of sagging through the rod **96**, thereby preventing a step difference generated between the two.

(96) Meanwhile, the suspender **SU** may include a cam **98** that is coupled to the other end of the rod **96** and is rotatable together with the rod **96**, and extends in the radial direction of the rod **96** to be in contact with the door **41**. In this case, the rod **96** and the cam **98** may be rotated in response to the rotation of the fastening member **97**.

(97) The cam **98** may include a first cam **98a** extending from the rod **96** by a first length D_a and a second cam **98b** extending from the rod **96** by a second length D_b . Here, the second length D_b may be longer than the first length D_a . In this case, in response to the rotation of the fastening member

97, the first cam **98a** or the second cam **98b** may be in contact with the door **41**, thereby adjusting the position of the door **41** with respect to the housing **30**.

(98) According to an aspect of the present disclosure, provided is a display device including: a flexible display panel; a roller which extends long, and around which the display panel is wound, or from which the display panel is unwound; a housing which has the roller located therein, and has an opening through which the display panel passes; and a door assembly which is coupled to the housing while being adjacent to the opening, and opens and closes the opening; wherein the door assembly includes: a door which opens and closes the opening; and a suspender located between the housing and the door, the suspender that has one end fixed to any one of the housing and the door, and has the other end movably coupled to the other of the housing and the door.

(99) In addition, according to another aspect of the present disclosure, the suspender may further include a pin that extends long, wherein the door assembly may further include: a fastening part to which one end of the pin is fixed; and a sliding part which opposes the fastening part with respect to the pin, and to which the other end of the pin is movably coupled, and wherein the fastening part may be formed at any one of the housing and the door, and the sliding part may be formed at the other of the housing and the door.

(100) In addition, according to another aspect of the present disclosure, the sliding part may further include a slot through which the pin passes, and which extends long in a direction intersecting with a length direction of the pin, wherein the pin may include: a body having one end fixed to the fastening part; a neck which is coupled to the other end of the body, and inserted into the slot; and a head which opposes the body with respect to the neck, and has a diameter greater than a gap of the slot.

(101) In addition, according to another aspect of the present disclosure, the sliding part may include: a first part formed by penetrating the sliding part; and a second part which protrudes from an inner surface of the first part, and forms the slot therein; wherein an outer surface of the neck may be in contact with an inner surface of the second part, and the head may overlap with the second part in a length direction of the pin.

(102) In addition, according to another aspect of the present disclosure, the display device may further include a sheet which is located between the second part and the neck, and formed of a plastic material.

(103) In addition, according to another aspect of the present disclosure, any one of the fastening part and the sliding part may be formed while being recessed outwardly from an inner surface of the door.

(104) In addition, according to another aspect of the present disclosure, the pin and the sliding part may be applied with a mutual magnetic attraction.

(105) In addition, according to another aspect of the present disclosure, the pin may further include a magnet located in the other end of the pin, wherein the sliding part further may include a magnetic substance facing the magnet in a length direction of the pin.

(106) In addition, according to another aspect of the present disclosure, the magnetic substance may extend as much as the door moves while opening and closing the opening.

(107) In addition, according to another aspect of the present disclosure, the magnetic substance may be located to correspond to a portion of a movement trajectory of the door facing the magnet.

(108) In addition, according to another aspect of the present disclosure, the magnet may be formed in a ball or roller shape.

(109) In addition, according to another aspect of the present disclosure, the display device may further include a sheet which is located between the magnet and the magnetic substance, and formed of a plastic material.

(110) In addition, according to another aspect of the present disclosure, the opening and the door may extend long in a length direction of the roller, wherein the suspender may further include a rod that has one end fixed to the housing, and extends long in a length direction of the door.

(111) In addition, according to another aspect of the present disclosure, the door assembly may further include a holder which is coupled to the door, and in which a hole into which the rod is inserted is formed, wherein the hole may extend long in a moving direction of the door.

(112) In addition, according to another aspect of the present disclosure, the suspender may further include a cam which is coupled to the other end of the rod and is rotatable together with the rod, and extends in a radial direction of the rod to contact the door, wherein the cam may include: a first cam extending from the rod by a first length; and a second cam extending from the rod by a second length greater than the first length.

(113) In addition, according to another aspect of the present disclosure, the suspender may further include a plurality of suspenders spaced apart from each other in a length direction of the roller.

(114) Certain embodiments or other embodiments of the invention described above are not mutually exclusive or distinct from each other. Any or all elements of the embodiments of the invention described above may be combined or combined with each other in configuration or function.

(115) For example, a configuration “A” described in one embodiment of the invention and the drawings and a configuration “B” described in another embodiment of the invention and the drawings may be combined with each other. Namely, although the combination between the configurations is not directly described, the combination is possible except in the case where it is described that the combination is impossible.

(116) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A display device comprising: a flexible display panel; a roller around which the flexible display panel is wound, or from which the flexible display panel is unwound; a housing which has the roller located therein, and has an opening through which the flexible display panel passes; and a door assembly which is coupled to the housing while being adjacent to the opening, and opens and closes the opening, wherein the door assembly comprises: a door which opens and closes the opening; a suspender comprising a pin and located between the housing and the door, the suspender having one end fixed to any one of the housing and the door, and having another end movably coupled to the other of the housing and the door, a fastening part which is formed at any one of the housing and the door, and to which one end of the pin is fixed; and a sliding part which is formed at the other of the housing and the door, and which is positioned opposite the fastening part with respect to the pin, and to which another end of the pin is movably coupled, wherein the sliding part comprises: a first part formed by penetrating the sliding part; and a second part protruding from an inner surface of the first part, and forming a slot through which the pin passes, wherein the slot extends in a direction intersecting with a length direction of the pin, wherein the pin comprises: a body having one end fixed to the fastening part; and a neck which is coupled to the other end of the body, and inserted into the slot, wherein an outer surface of the neck is in contact with an inner surface of the second part, and wherein the display device further comprises a sheet which is located between the second part and the neck, and formed of a plastic material.

2. The display device of claim 1, wherein the pin further comprises: a head which is opposite to the body with respect to the neck, and which has a diameter greater than a gap of the slot.

3. The display device of claim 2, wherein the head overlaps with the second part in a length direction of the pin.

4. The display device of claim 1, wherein any one of the fastening part and the sliding part is formed while being recessed outwardly from an inner surface of the door.

5. A display device comprising: a flexible display panel; a roller around which the flexible display panel is wound, or from which the flexible display panel is unwound; a housing which has the roller located therein, and has an opening through which the flexible display panel passes; and a door assembly which is coupled to the housing while being adjacent to the opening, and opens and closes the opening; wherein the door assembly comprises: a door which opens and closes the opening; and a suspender comprising a pin and located between the housing and the door, the suspender having one end fixed to any one of the housing and the door, and having another end movably coupled to the other of the housing and the door; a fastening part which is formed at any one of the housing and the door, and to which one end of the pin is fixed; and a sliding part which is formed at the other of the housing and the door, and which is positioned opposite the fastening part with respect to the pin, and to which the other end of the pin is movably coupled, and which comprises a magnetic substance, wherein the pin comprises a magnet located at the other end of the pin, which faces the magnetic substance in a length direction of the pin, and which is formed in a ball or roller shape, and wherein the pin and the sliding part are applied with a mutual magnetic attraction.

6. The display device of claim 5, wherein the magnetic substance extends as much as the door moves while opening and closing the opening.

7. The display device of claim 5, wherein the magnetic substance is located to correspond to a portion of a movement trajectory of the door facing the magnet.

8. The display device of claim 5, further comprising a sheet which is located between the magnet and the magnetic substance, and formed of a plastic material.

9. A display device comprising: a flexible display panel; a roller around which the flexible display panel is wound, or from which the flexible display panel is unwound; a housing which has the roller located therein, and has an opening through which the flexible display panel passes; and a door assembly which is coupled to the housing while being adjacent to the opening, and opens and closes the opening; wherein the opening extends in a length direction of the roller, wherein the door assembly comprises: a door which opens and closes the opening, and which extends in a length direction of the roller; a suspender located between the housing and the door, the suspender having one end fixed to any one of the housing and the door, and having another end movably coupled to the other of the housing and the door, and the suspender comprises a rod that has one end fixed to the housing, and extends in a length direction of the door; and a holder which is coupled to the door, and in which a hole into which the rod is inserted is formed, wherein the hole extends in a moving direction of the door, wherein the suspender further comprises a cam which is coupled to the other end of the rod and is rotatable together with the rod, and extends in a radial direction of the rod to contact the door, and wherein the cam comprises: a first cam extending from the rod by a first length; and a second cam extending from the rod by a second length greater than the first length.

10. The display device of claim 1, wherein the suspender further comprises a plurality of suspenders spaced apart from each other in a length direction of the roller.
